

THE \$60 CHIP THAT CHARGES YOUR PHONE OFF A WOOD STOVE

A flat ceramic square turns stove heat into electricity, with no sun and no grid. The catch is keeping the top cool.

HOW THE CHIP WORKS

It is a flat ceramic square about 2 inches across with two wires. Heat the bottom, keep the top cool, and it makes electricity. The chip does not make power from heat. It makes power from a difference in heat.

Inside is a grid of tiny bismuth telluride pillars. Heat one end and the charges pile up at the cold end, leaving a voltage. The build chip has 126 pairs of pillars in a 56 millimeter square.

8.4 volts

Open circuit at 300 Celsius on one face and 30 on the other, off the maker's sheet.

14 watts

With a load, on the sheet, if the top face stays at 30 Celsius.

a few watts

What to really plan on. A fan cooled sink on a stove cannot hold the top that cold, so the gap shrinks and power drops.

SHOPPING LIST

- ☐ **The chip.** A 56 millimeter bismuth telluride generator module rated for a 300 Celsius hot side. About 56 dollars from the maker.
- ☐ **Heat sink with a fan.** An old desktop computer cooler works. Swap its 12 volt fan for a 5 volt fan of the same size, a few dollars.
- ☐ **Voltage regulator.** A small buck boost board that takes about 3 volts up to 20 or more and holds 5 volts out. Set the output to 5 with a meter first.
- ☐ **Heat transfer layer** the chip maker recommends. If it comes with graphite sheets, do not add paste on top.
- ☐ **Steel base plate and two bolts.** The chip sits between the plate and the heat sink.
- ☐ **Heat rated wire,** silicone or fiberglass insulated, for the leads.
- ☐ **Stove thermometer,** to find and hold your spot.

WHAT IT COSTS VS WHAT YOU CAN BUY

under \$100

The whole build at full retail.

\$200

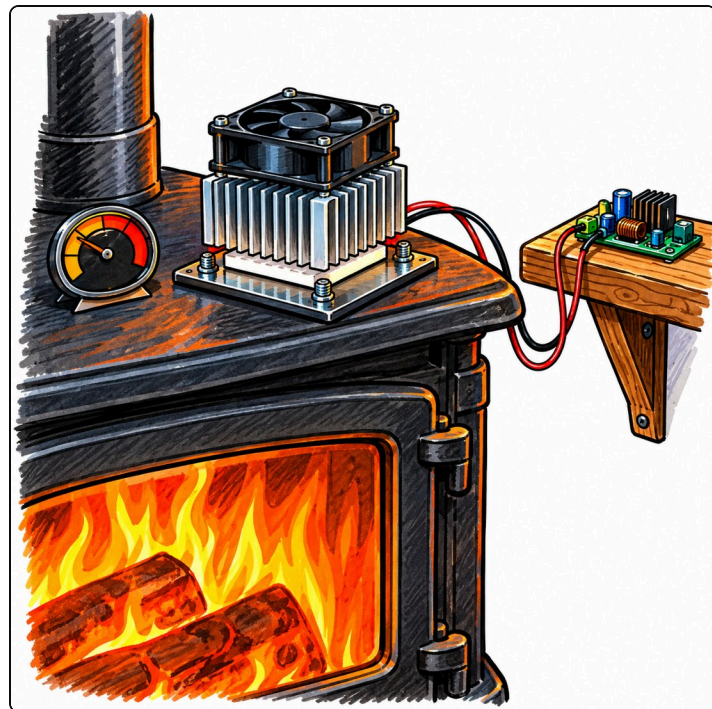
A BioLite camp stove with a charging port. It peaks at 3 watts.

\$165 to \$200

Caframo's top Ecofan model. Same kind of chip, spinning a fan blade instead of filling a charging port. Caframo says the module lasts about 10 years.

\$600

A Jackery 300 with a 100 watt panel, or a small solar and battery kit.

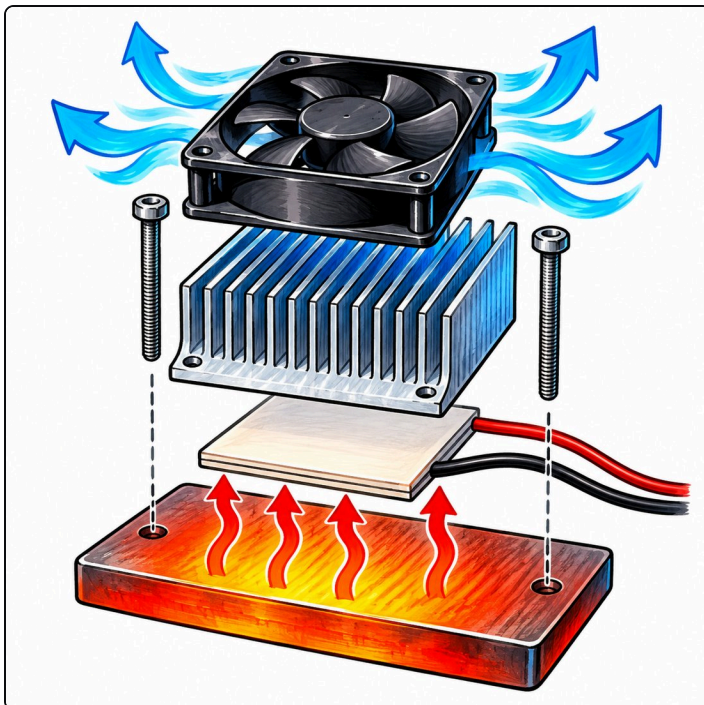


Chip under a fan cooled heat sink on a steel plate, thermometer beside it, regulator mounted off the stove. Same layout as an Ecofan.

On a bright day the solar kit wins easily. Under heavy cloud, panels make 10 to 25 percent of their rating. Under snow, close to nothing. At night, nothing. A wood stove in a winter outage runs hottest at night, exactly when a panel is useless.

THE BUILD, IN ORDER

- 1 Stack it. The chip sits between the steel base plate and the heat sink. Run the two bolts through the sink into the plate.
- 2 Snug the bolts down evenly, a little at a time on each side. Uneven pressure cracks chips.
- 3 Add the heat transfer layer the maker recommends between every surface. Any air gap blocks heat, and blocked heat is lost power.
- 4 Wire the chip leads to the regulator input. Red to positive, black to negative.
- 5 Run the 5 volt fan and the charging port off the regulator output.
- 6 Use heat rated wire. Route it up and away from the plate, and mount the regulator off the stove, a foot or more away.



Bottom to top: steel base plate, chip, heat sink with fan, bolts through. Even, gentle pressure only.

WANT MORE POWER?

Put a second chip beside the first and wire them in series. The voltage adds up, and a regulator that takes 20 volts in still handles it. Two chips is a sensible ceiling for a fan cooled build.

TEST IT BEFORE YOU NEED IT

- ☐ Build it in the fall and try it on a normal evening. Do not let the first test be the night the power goes out.
- ☐ Before it touches the stove, warm the base plate on a hot plate set low. Put a multimeter on the chip's two leads.
- ☐ The voltage should climb past a volt, then two, then more. If it sits near zero, check your meter leads, then the contact layers, then tighten the bolts evenly.
- ☐ If the voltage climbs but the regulator stays dark, check the polarity. Red goes to the positive input.
- ☐ If the regulator lights but the fan will not spin, check that it is a 5 volt fan wired to the output.
- ☐ Once it works on the burner, move it to the stove. Find the spot that holds around 450 to 500 Fahrenheit on a normal fire. Mark it with a dab of stove paint.
- ☐ Run it a whole evening and see what lands in the power bank. Now you know your real number.

THE HEAT LIMIT

The chip is rated for a 300 Celsius hot face, which is 572 Fahrenheit. Keep the stove thermometer beside it under about 550 Fahrenheit, and set the chip toward a cooler edge, not the hottest spot. Watch out for cheap 40 millimeter chips, often rated only to 150 Celsius, which is 302 Fahrenheit. A wood stove kills those fast. Most stove top thermometers mark a best burn zone of about 400 to 600 Fahrenheit, which runs past the chip's limit. Check the maximum hot side temperature before you buy anything.

RUNNING IT

- ☐ Set the base plate on the cooler edge of the stove top, thermometer beside it. Let the fire settle. The regulator light comes on and the fan starts turning.
- ☐ Charge a power bank first, not the phone. A fire rises and falls, and so does the chip's output. The bank soaks up the uneven power, and the phone charges smoothly from the bank.
- ☐ If the thermometer climbs past 550 Fahrenheit, slide the plate farther toward the edge.
- ☐ Keep the phone and the power bank off the stove.

THE CHARGE MATH

14 Wh	Roughly what a phone like an iPhone 16 holds.
about 1/3	What you lose between the regulator and the power bank.
20 Wh	Chip output for one full phone charge.
5 hours	At 4 watts, that is about one long evening of fire per phone.

WHAT IT RUNS, AND WHAT IT WILL NOT

Runs: a phone, a small plug in lamp at 1 to 3 watts, many weather radios, a headlamp, a small fan, most walkie talkies. A trickle that never stops while the fire burns.

Will not: anything that makes heat or cold. A space heater, a fridge, a microwave, or a hair dryer each want hundreds or thousands of watts.

WHY IT MATTERS

In September 2024, Hurricane Helene hit western North Carolina. Two days after landfall, the Federal Communications Commission reported 74 percent of cell sites in its North Carolina reporting area out of service.

Two weeks after landfall, Duke Energy still had more than 80,000 customers in the Carolinas without power. Every time a tower came back, it only helped the people whose phones still had a charge.

Federal guidance: keep your phone charged, and text instead of calling. A text can squeeze through a jammed network when a call will not. A 20 dollar power bank is great on night one and empty on day nine. The chip refills it every evening the fire is lit.

SAFETY RULES. NOT OPTIONAL.

- ☐ This is a hands on project with real heat and real wiring. Never leave it running with nobody home.
- ☐ The plate gets hot enough to burn you badly. Handle it with gloves.
- ☐ Keep the stove's clearances, and keep a working carbon monoxide alarm in the room.
- ☐ Set the regulator's output to 5 volts with a meter before anything touches a phone.



The stove fills the power bank while the fire is hot. The bank carries you through the day.

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